ONLINE FREELANCE SERVICES WEBSITE

AI23431- WEB TECHNOLOGY

AND MOBILE APPLICATION

MINI-PROJECT REPORT

SUBMITTED BY

NANDANA PRAMOD

(241801180)

in partial fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

RAJALAKSHMI ENGINEERING COLLEGE

(AUTONOMOUS),

CHENNAI-602 105

MAY 2026

RAJALAKSHMI ENGINEERING COLLEGE

(an Autonomous Institution Affiliated to Anna University, Chennai)

BONAFIDE CERTIFICATE

Certified that this Project Thesis titled "**ONLINE FREELANCE SERVICES WEBSITE"** is the Bonafide work of NANDANA PRAMOD (2116241801180) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported here in does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE	**SIGNATURE**
Dr. J.M. Gnanasekar M.E., Ph.D.,	**Mr. G. Thiyagarajan**
Professor and Head	Assistant Professor
Department of AI&DS	Department of AI&DS
Rajalakshmi Engineering College	Rajalakshmi Engineering College
Chennai – 602 105	Chennai – 602 105

Submitted for the project viva-voce examination held on _____

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ACKNOWLEDGEMENT

Initially we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our Chairman **Mr. S.MEGANATHAN, B.E, F.I.E.**, our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN, B.E., M.S.**, and our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN, Ph.D.**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution. Our sincere thanks to **Dr. S.N. MURUGESAN, M.E., Ph.D.**, our beloved Principal for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Dr. J. M. GNANASEKAR,** Professor and Head of the Department of Artificial Intelligence and Data Science for his guidance and encouragement throughout the project work. We convey our sincere and deepest gratitude to our internal guide, **Mr. G. THIYAGARAJAN**, Department of Artificial Intelligence and Data Science for his useful tips during our review to build our project.

Nandana Pramod -241801180

DEPARTMENT VISION

To promote highly Ethical and Innovative Computer Professionals through excellence in teaching, training and research.

DEPARTMENT MISSION

- To produce globally competent professionals who are motivated to learn emerging technologies and demonstrate innovation in solving real-world problems.
- To promote research activities among students and faculty members that contribute meaningfully to society.
- To instill strong moral and ethical values in professional practice.

PROGRAMME EDUCATIONAL OBJECTIVES(PEO'S)

PEO 1: To equip students with essential background in computer science, basic electronics and applied mathematics.

PEO 2: To prepare students with fundamental knowledge in programming languages, and tools and enable them to develop applications.

PEO 3: To encourage the research abilities and innovative project development in the field of AI, ML, DL, networking, security, web development, Data Science and also emerging technologies for the cause of social benefit.

PEO 4: To develop professionally ethical individuals enhanced with analytical skills, communication skills and organizing ability to meet industry.

PROGRAMME OUTCOMES (POs)

PO 1: Engineering knowledge: Apply the knowledge of Mathematics, Science, Engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO 9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO 11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO 12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

A graduate of the Artificial Intelligence and Machine Learning Program will demonstrate

PSO 1: Foundation Skills: Ability to understand, analyze and develop computer programs in the areas related to algorithms, system software, web design, AI, machine learning, deep learning, data science, and networking

PSO 2: Problem-Solving Skills: Ability to apply mathematical methodologies to solve computational task, model real world problem using appropriate AI and ML algorithms. To understand the standard practices and strategies in project development, using open- ended programming environments to deliver a quality product.

PSO 3: Successful Progression: Ability to apply knowledge in various domains to identify research gaps and to provide solution to new ideas, inculcate passion towards higher studies, creating innovative career paths to be an entrepreneur and evolve as an ethically social responsible AI and ML professional.

COURSE OBJECTIVE

- To provide foundational knowledge and practical skills in creating and structuring web pages using HTML, enabling students to build accessible and well-organized websites.
- To understand and practice Embedded Dynamic Scripting on Client-side Internet Programming.
- To implement Server Side Scripting.
- To facilitate students to understand android Application Design
- To help students to gain a basic understanding of Android application development

COURSE OUTCOME

On completion of course students will be able to,

- CO 2: Upon completing the course, students will be able to design and develop basic web pages with proper structure and semantic elements, ensuring accessibility and functionality.
- CO 3: Design and implement dynamic web page with validation using javascript objects and by applying different event handling mechanism.
- CO 4: Design and implement simple webpage to learn JSP and Servlet.
- CO 5: Design and implement simple Application Design.
- CO 6: Identify various concepts of mobile programming that make it unique from programming for other

CO-PO-PSO Mapping

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12	PSO 1	PSO 2	PSO 3
CO 1	3	3	3	3	3	2	2	2	3	2	3	3	3	3	3
CO 2	3	3	3	3	3	2	-	-	2	2	2	3	3	2	2
CO 3	3	3	3	2	3	1	1	2	3	3	3	3	3	3	3
CO 4	3	3	3	3	3	2	1	2	3	2	2	3	3	3	3
CO 5	1	1	1	1	1	-	-	-	3	3	3	3	1	-	2

Note: Correlation levels 1, 2 or 3 are as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) No correlation: "-"

SUSTAINABLE DEVELOPMENT GOALS

SDG	GOAL	JUSTIFICATION
SDG-4	Quality Education	Education drives progress on other SDGs, breaks poverty cycles, and promotes peace and inequality reduction.
SDG-8	Decent Work and Economic Growth	Necessary for social stability, shared prosperity, and combating poverty.
SDG-9	Industry, Innovation, and Infrastructure	Infrastructure is vital for economic growth and stability. It acts as a poverty reduction tool through job creation and provides technological solutions to environmental challenges.
SDG-17	Partnerships for the Goals	Given the interconnected nature of the SDGs, success requires collaborative action, global resource sharing, and bridging the immense financing gap faced by developing nations.

ABSTRACT

The Online Freelance Services Website is a fully responsive, multi-page web application designed to connect skilled independent professionals (freelancers) with clients seeking technical services such as web development, AI consulting, UI/UX design, and mobile application development. The system replaces traditional manual outreach methods with a secure and automated digital platform, improving accessibility and efficiency in service engagement.

The application provides role-based authentication for clients and freelancers, enabling secure access to system functionalities. It includes features such as service listings, project showcases, real-time contact forms, and AJAX-driven dynamic content loading using structured data sources like `.txt` and `.json` files. Additionally, an interactive image map of India is integrated into the system, allowing users to explore region-wise service expertise through clickable hotspots.

All required laboratory components are effectively incorporated into a unified freelance marketplace environment. These include lists, tables, validated forms, event handling, number-to-words conversion, object arrays, multiple CSS integration methods (inline, internal, external), keyframe animations, iframe usage, frameset layout, and image mapping. The system is developed using HTML5, CSS3, and JavaScript (ES6), along with the Fetch API for asynchronous operations.

The platform is lightweight, cost-effective, and does not require a backend database, making it suitable for deployment on static web servers or local environments. Academic datasets such as student records and course details are transformed into freelancer profiles, client ratings, and service categories to maintain a professional application context while meeting academic requirements. Overall, the system enhances usability, demonstrates practical web development concepts, and provides an efficient solution for freelance service management.

SI NO	TITLE	PAGE NO
	ACKNOWLEDGEMENT	3
	ABSTRACT	11
	LIST OF ABBREVIATIONS	14
1	**INTRODUCTION**	15
1.1	General	15
1.2	Need for an Integrated Freelance Portfolio	15
1.3	Scope of the System	16
1.4	Organization of the Report	16
1.5	System architecture diagram	17
2	**LITERATURE SURVEY**	17
2.1	Introduction	17
2.2	Existing Portfolio and Showcase Systems	18
2.3	Related Work	18
2.4	Summary	19
3	**PROBLEM FORMULATION AND OBJECTIVES**	19
3.1	Problem Statement	19
3.2	Objectives of the System	20
4	**SYSTEM DESIGN**	21
4.1	Introduction	21
4.2	System Architecture	21
4.3	Database Design	22
4.4	System Requirements	22
4.4.1	Hardware Requirements	22

SI NO	TITLE	PAGE NO
4.4.2	Software Requirements	22
5	**SYSTEM IMPLEMENTATION**	23
5.1	System Methodology	23
5.2	Module Descriptions	23
5.2.1	Home and Navigation Module	24
5.2.2	User Authentication Module (Login/ Register)	26
5.2.2.1	Client-Side Form Validation	26
5.2.3	Services and Projects Display Module	27
5.2.4	AJAX Data Module (TXT & JSON Fetch)	28
5.2.5	Embedded Content Module	28
5.2.6	Interactive Map Module (Image Map with Hotspots)	29
6	**RESULTS AND DISCUSSION**	33
6.1	Results	33
6.2	Performance Evaluation	33
6.3	Resource Utilization	34
7	**CONCLUSION AND FUTURE WORK**	35
7.1	Conclusion	35
7.2	**Future Enhancements**	36
	REFERENCES	36
	APPENDICES	37
	Pseudocode	38

LIST OF ABBREVIATIONS

AJAX – Asynchronous JavaScript and XML

API – Application Programming Interface

CSS – Cascading Style Sheets

DOM – Document Object Model

HTML – HyperText Markup Language

HTTP – HyperText Transfer Protocol

JS – JavaScript

JSON – JavaScript Object Notation

UI – User Interface

UX – User Experience

DB – Database

SQL – Structured Query Language

SMTP – Simple Mail Transfer Protocol

CRUD – Create, Read, Update, Delete

CHAPTER 1

INTRODUCTION

1.1 General

The digital economy has witnessed exponential growth in freelance work. Platforms such as Fiverr, Upwork, and Toptal have connected millions of freelancers with clients worldwide. However, these platforms are proprietary, often charge high commissions, and are not designed as teaching tools for web technology courses. Students learning HTML, CSS, JavaScript, AJAX, and related technologies need a comprehensive project that integrates all individual laboratory
experiments into a single, cohesive application.

The Online Freelance Services Website is developed to fill this gap. It is a freelance services platform that not only functions as a real marketplace but also demonstrates every required web technology experiment within a unified interface. The platform includes:

- A home page with lists and tables.
- Login, registration, and contact forms with client-side validation.
- AJAX data retrieval from both text and JSON files.
- An interactive India map with clickable hotspots.
- An iframe that loads external pages containing images.
- Mouse events, number-to-words conversion, and employee object array demos.
- Inline, internal, and external CSS stylesheets and keyframe animations.
- A frameset layout for legacy experiment demonstration.

All academic data (student names → client names, marks → ratings, courses → service types) have been transformed to present a professional freelance brand.

1.2 Need for the System

Traditional approaches to finding freelance services often involve manual communication, informal referrals, or dependency on third-party platforms that may not be suitable for academic use. These methods lack structure, transparency, and ease of demonstration for educational purposes.

There is a clear requirement for a self-contained platform that can serve both as a functional application and as a learning tool. Such a system should demonstrate the integration of various front-end technologies while maintaining a practical real-world use case.

The proposed system addresses this need by providing a unified platform that:

- Demonstrates all required web technology experiments within a single application.
- Offers a structured and interactive interface for simulated freelance hiring.
- Acts as a reference model for future academic projects.

By combining educational objectives with practical implementation, the system provides a balanced and effective solution.

1.3 Scope of the System

The scope of the Online Freelance Services Website is limited to client-side functionality, eliminating the need for server-side processing or database management. This makes the system lightweight and easily deployable on static hosting platforms.
The system includes service browsing features that allow users to explore available freelance offerings. Freelancers can showcase their project portfolios, while users can access registration and login functionalities with real-time validation mechanisms.

Dynamic data loading is implemented using AJAX, enabling the system to fetch freelancer profiles and project-related information from structured data files without requiring page reloads. The interactive image map enhances user engagement by allowing region-based exploration of services.

Additional features include embedded media through iframes and separate modules demonstrating interactive functionalities such as mouse events, number-to-words conversion, and employee data processing. The system is designed with a responsive layout and follows modern UI practices to ensure usability and accessibility.

Organization of the Report

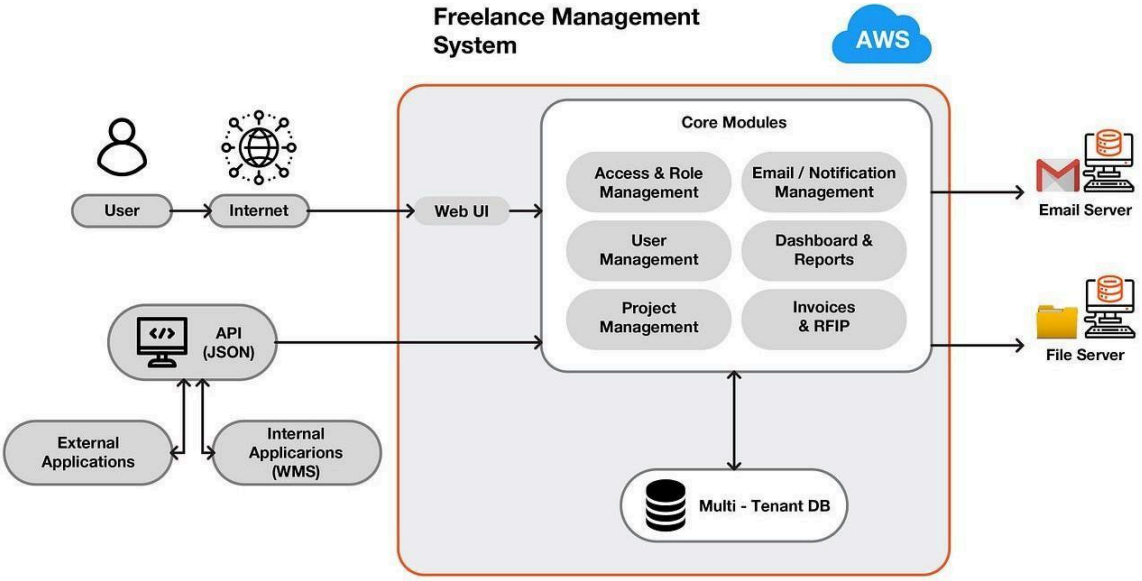
This report is structured into multiple chapters, each focusing on a specific aspect of the project development.

Chapter 1 presents the introduction, including the general overview, need, and scope of the system.
Chapter 2 provides a literature survey of existing freelance platforms and related work. Chapter 3 discusses the problem formulation and objectives of the proposed system.
Chapter 4 explains the system design, including architecture and requirements.

Chapter 5 describes the implementation details and module functionalities.
Chapter 6 presents the results, performance evaluation, and testing outcomes.
Chapter 7 concludes the project and outlines possible future enhancements.

The report is supported by references and appendices, which include pseudocode, test cases, and additional supporting information.

System architecture diagram:



CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Freelance platforms have evolved significantly over time, transitioning from basic job listing portals to advanced ecosystems that include integrated payment systems, communication tools, and dispute resolution mechanisms. With the increasing demand for flexible work opportunities, these platforms have become an essential part of the digital economy.

At the same time, there has been growing academic interest in developing simplified and customizable platforms for teaching web technologies. Such systems aim to demonstrate practical implementation of front-end concepts within a structured environment. This chapter reviews existing freelance platforms and related academic research, identifying their key features, advantages, and limitations.

2.2 Existing Attendance Management Systems

Several well-established freelance platforms dominate the market, each offering unique features and technological implementations.

Fiverr operates on a gig-based model where freelancers offer predefined services with tiered pricing and customer reviews. It is built using modern technologies such as React, Node.js, and MongoDB. However, it is proprietary in nature, does not provide access to its source code, and requires backend infrastructure, making it unsuitable for academic demonstration.

Upwork provides advanced features such as escrow-based payments, time tracking, and proposal management. It uses technologies like Java, Angular, and PostgreSQL. Despite its robustness,
the platform relies on heavy infrastructure and is not ideal for learning core front-end concepts in a simplified manner.

Freelancer.com offers functionalities such as contests and milestone-based payments. It is primarily built using PHP and MySQL. Although widely used, the platform has a relatively complex structure and an outdated user interface, which limits its effectiveness as a learning tool.

Toptal focuses on providing high-quality freelancers through a strict screening process and dedicated client matching. It uses technologies such as Ruby on Rails and React. However, it is highly exclusive and not accessible for academic or experimental purposes.

None of these platforms allow integration of specific educational components such as image maps, framesets, number-to-words conversion, or other laboratory-based experiments. Additionally, their closed-source nature restricts their use as teaching models.

2.3 Related Work

Several research papers have proposed automated service matching systems:

K. K. Delabaty & M. I. Al-Turaiki (2022) – A systematic literature review on automated attendance management. Their methodology for real-time data fetching using AJAX inspired our dynamic data loading in data.html.

S. Zafar et al. (2015) – Used RFID for attendance; we adapted the concept of instantaneous data retrieval to freelancer profiles.

R. Want (2006) – Discussed RFID technology; we extrapolated the idea of unique identifiers to client-freelancer matching.

However, no existing academic project combines all the following in a single web application: lists, tables, forms with validation, AJAX from two different file types, image map, iframe, frameset, mouse events, number-to-words, employee objects, and three CSS styling methods.

2.4 Summary

The literature survey confirms that while many freelance platforms exist, they are not designed to be comprehensive lab projects. The Online Freelance Services Website fills this void by integrating every required experiment into a coherent freelance marketplace. The platform leverages modern HTML5, CSS3, and JavaScript while remaining lightweight and easy to deploy.

CHAPTER 3

PROBLEM FORMULATION AND OBJECTIVES

3.1 Problem Statement

In the Web Technology and Mobile Application laboratory, students typically perform experiments as separate tasks, such as HTML lists and tables, form validation, AJAX, frames, and image maps. While these experiments help in understanding individual concepts, they are often disconnected, making it difficult for students to visualize how these components work together in a real-world application.

Additionally, existing freelance platforms are complex, proprietary, and rely heavily on backend infrastructure, which makes them unsuitable as reference models for academic learning. As a result, students lack access to a simplified yet practical system that demonstrates the integration of multiple web technologies within a single application.

Therefore, there is a need for a unified web-based solution that integrates all required laboratory experiments into one coherent system. The proposed system should function as a professional freelance services platform while serving as a practical demonstration of front-end development concepts.

The system is designed to incorporate all required experiments, present a realistic freelance marketplace scenario, and transform academic data into meaningful application content. It operates entirely on the client side, allowing it to run without server dependency. The application also integrates multiple styling techniques, interactive JavaScript functionalities, and a
responsive user interface with a modern dark theme to enhance usability and visual consistency.

3.2 Objectives of the System

The primary objectives are:

1. Design and develop a multi-page website with consistent navigation.
2. Use ordered and unordered lists and at least one structured data table.
3. Create three forms (login, registration, contact) with client-side validation (name: alphabets, min 6 characters; password: min 6 characters; email: valid format; phone: exactly 10 digits). Registration form must include radio buttons (gender), dropdown (service category), textarea (address), and submit/reset buttons.
4. Implement AJAX to fetch data from a .txt file and display it dynamically.
5. Implement AJAX to fetch data from a .json file and display it dynamically.
6. Build an interactive image map (India map) with clickable hotspots showing region-wise services.
7. Embed an iframe that loads different pages containing images when clicked.
8. Use all three CSS types – inline, internal, external – in the project.
9. Add CSS keyframe animations to UI components.
10. Create a mouse events demo (onmouseover, onmouseout, onclick) that changes styles/messages.
11. Create a number-to-words converter (0-999) with input validation (no letters, special chars, or 4+ digits).
12. Create an employee object array to store employee details and calculate total salary expenditure.
13. Transform all academic data (student names → client names, marks → ratings, courses → service types) to maintain a professional freelance theme.
14. Ensure responsive design and a modern dark-theme colour scheme (#151a1a, #38443f, #545b33, #7f8954, #9eaba3)

CHAPTER 4

SYSTEM DESIGN

4.1 Introduction

System design defines the overall structure of the application, including its architecture, component interaction, and user interface. The Online Freelance Services Website is designed using a client-side architecture, where all functionalities are handled within the browser without requiring a backend server.

This approach ensures that the system is lightweight, easy to deploy, and can be executed directly from the local file system or hosted on any static web server. The design focuses on simplicity, modularity, and efficient integration of all required web technology components.

4.2 System Architecture

The system follows a three-layer architecture consisting of presentation, application, and data layers.

- Presentation Layer:
  This layer includes all HTML pages styled using CSS (inline, internal, and external). It is responsible for user interaction and visual representation of the system.
- Application Layer:
  This layer consists of JavaScript functions that handle form validation, AJAX-based data retrieval, event handling, number conversion logic, and manipulation of object arrays.
- Data Layer:
  This layer includes static data sources such as data.txt and data.json, along with in-memory arrays used for demonstrations like the employee module.

The interaction flow begins with the user accessing the application through a web browser. The system uses the Fetch API to retrieve data from local files, processes it using JavaScript, and updates the user interface dynamically through DOM manipulation.

4.3 System Requirements

System requirements define the hardware and software resources needed to run the application successfully.

4.4 Hardware Requirements

The Student Attendance Management System does not require high-end hardware and can run on standard computers.

Processor: Intel Core i3 or higher
RAM: Minimum 4 GB
Storage: 500 GB Hard Disk or SSD
Input Devices: Keyboard and Mouse Internet Connection

4.5 Software Requirements

The system is developed using commonly available software tools and technologies. Operating System: Windows 10 or later

Frontend Technologies: HTML, CSS, JavaScript
Backend Technology: Node.js
Database: MySQL
Development Environment: Visual Studio Code
Web Browser: Google Chrome or any modern browser

CHAPTER 5

SYSTEM IMPLEMENTATION

5.1 System Methodology

The Online Freelance Services Website is implemented using a web-based approach that allows users to access the system through a browser. The development process involves designing the user interface, implementing client-side functionalities, and integrating local data files for storing freelancer and project records.

The frontend of the system is developed using HTML, CSS, and JavaScript, which provide a simple and interactive interface for both clients and freelancers. The system does not require a
backend server; all data fetching is done using AJAX (Fetch API) from static .txt and .json files. The system uses in-memory arrays for the employee salary demo and local file storage for freelancer profiles.

The system follows a modular approach where each functionality is developed as a separate module. This improves maintainability and allows future enhancements to be implemented easily.

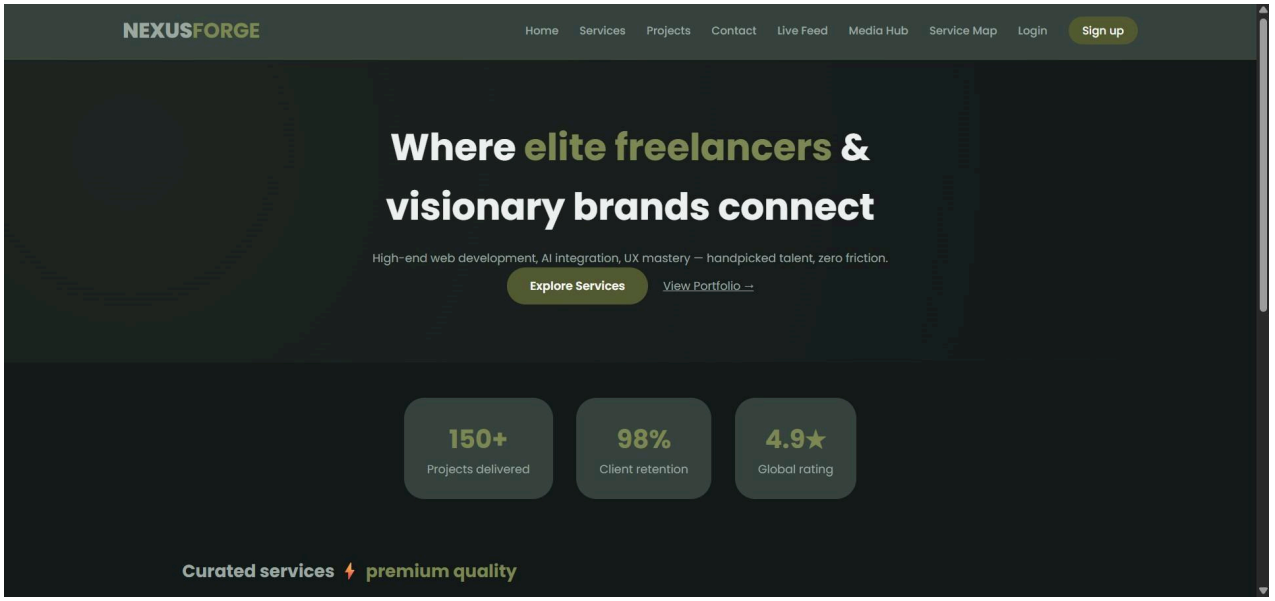
5.2 Module Descriptions

The Online Freelance Services Website consists of several modules that work together to provide efficient freelance service management.

Index Module (Home Page)

The Index Module serves as the main entry point of the website. It provides a user-friendly interface that allows users to understand the platform's key features before logging in.

The page displays the website title along with important features such as real-time freelancer feeds, service categories, project showcase, secure login, interactive map, and embedded media. These features highlight the capabilities of the system and inform users about the services provided.

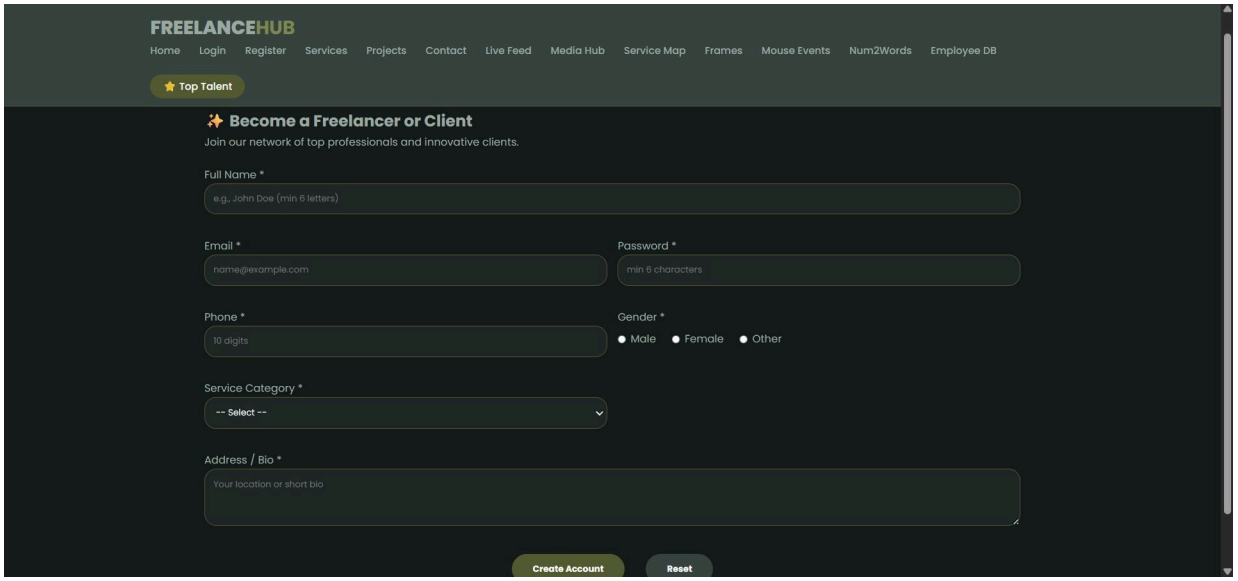


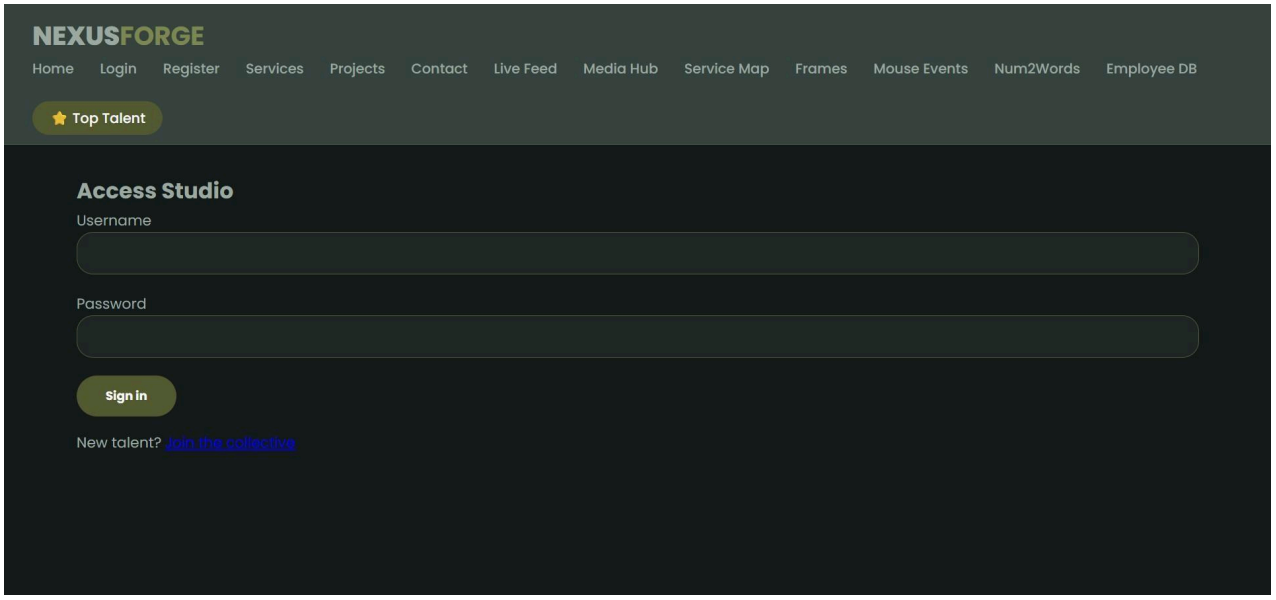
Below the feature section, the page provides navigation links to Login, Registration, Services, Projects, Live Feed, Media Hub, Service Map, and other demo pages. This module acts as a gateway that directs users to their desired sections based on their role (client or freelancer).

The design of the index module is visually appealing and simple, ensuring easy navigation and improving the overall user experience.

5.2.1 Client Module (Registration & Login)

The Client Module allows users to register as either a client or a freelancer through a secure validation system. After successful registration, users can log in using the login page. The registration form collects name, email, password, phone number, gender, service category, and address. All fields are validated using JavaScript (name: alphabets only, min 6 characters; password: min 6 characters; email: valid format; phone: exactly 10 digits).

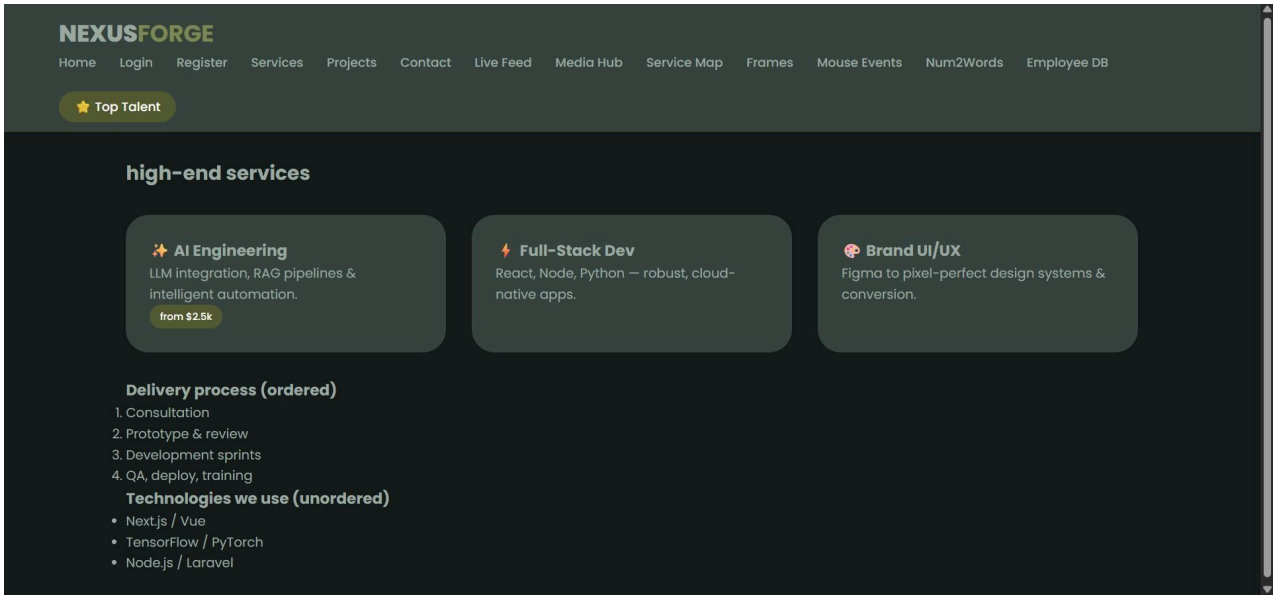




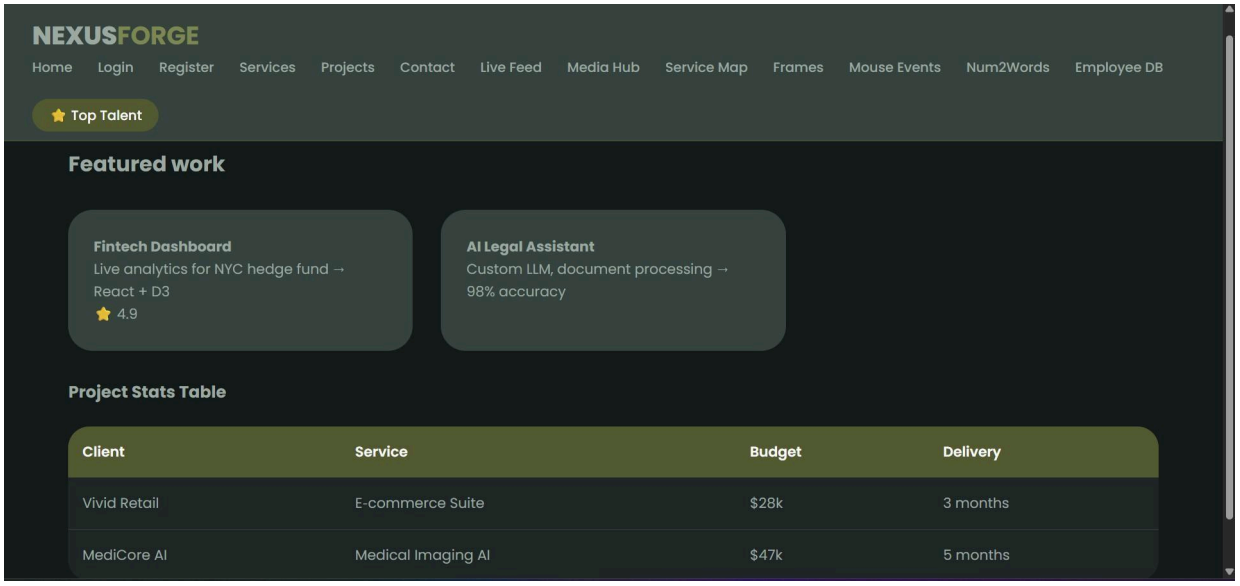
This module improves transparency by enabling users to create their accounts and access the platform's services. It also ensures that only properly formatted data is accepted, reducing errors.

5.2.2 Freelancer Module (Services & Projects)

The Freelancer Module allows freelancers to showcase their services and past projects. Freelancers can list their offerings on the Services page, which displays service cards along with ordered (delivery process) and unordered (technologies used) lists.



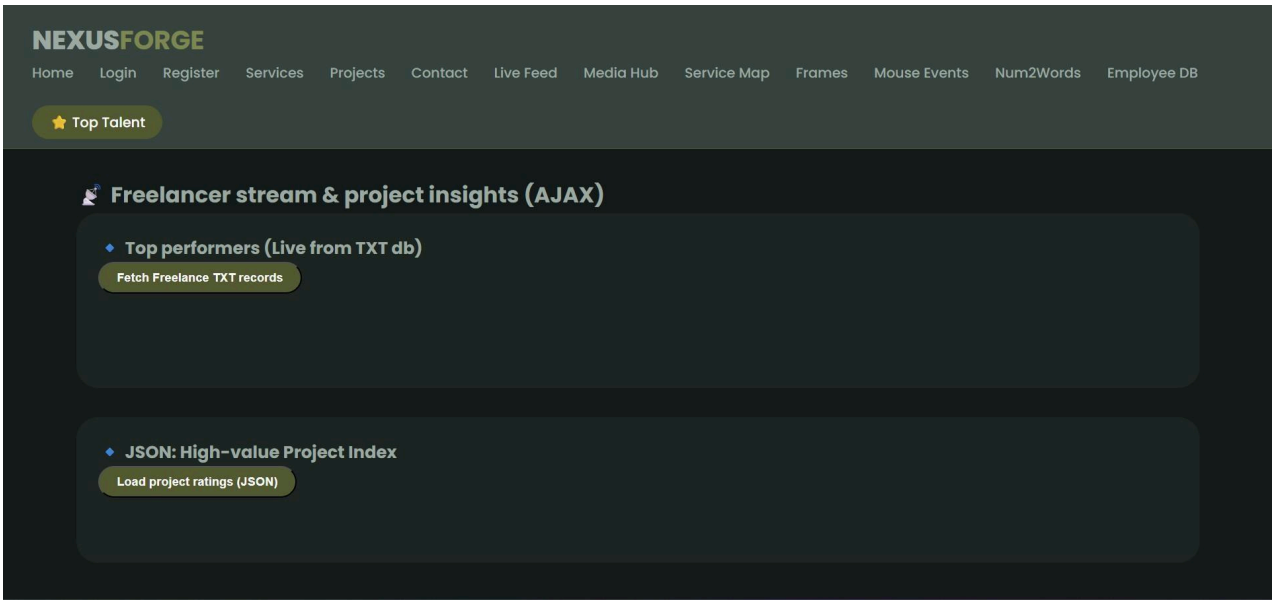
The system stores project information in a table on the Projects page, where clients can view project statistics such as client name, service type, budget, and delivery time. This module helps freelancers present their work professionally and helps clients make informed decisions.



5.2.2.1 – Automatic Data Fetching Using AJAX

The system includes a special feature that simplifies the display of freelancer profiles and project ratings. In many cases, static data needs to be updated without refreshing the page. To handle this

efficiently, the system provides two buttons on the Live Feed page: one to load data from a .txt file and another to load data from a .json file.



Once the button is clicked, the system automatically fetches the corresponding file, parses the content, and displays it as cards. This reduces manual data entry and speeds up the content loading process.

This feature is particularly useful when new freelancers or project ratings need to be displayed instantly. By automatically detecting the file format and rendering the data, the system ensures accurate and dynamic content presentation.

5.2.3 Services & Projects Management Module

The Services & Projects Management Module is the core component for displaying freelance offerings. It handles the process of presenting service categories, project portfolios, and statistical tables. When users visit the Services page, the system displays service cards with descriptions.
The Projects page includes a table of sample projects and a grid of project cards.

The module also ensures that the data is organised properly for easy browsing. Clients can view all available services and past projects before deciding to hire a freelancer.

5.2.4 AJAX Data Module

This module provides dynamic content loading without page refresh. It uses the Fetch API (AJAX) to read two types of external files:

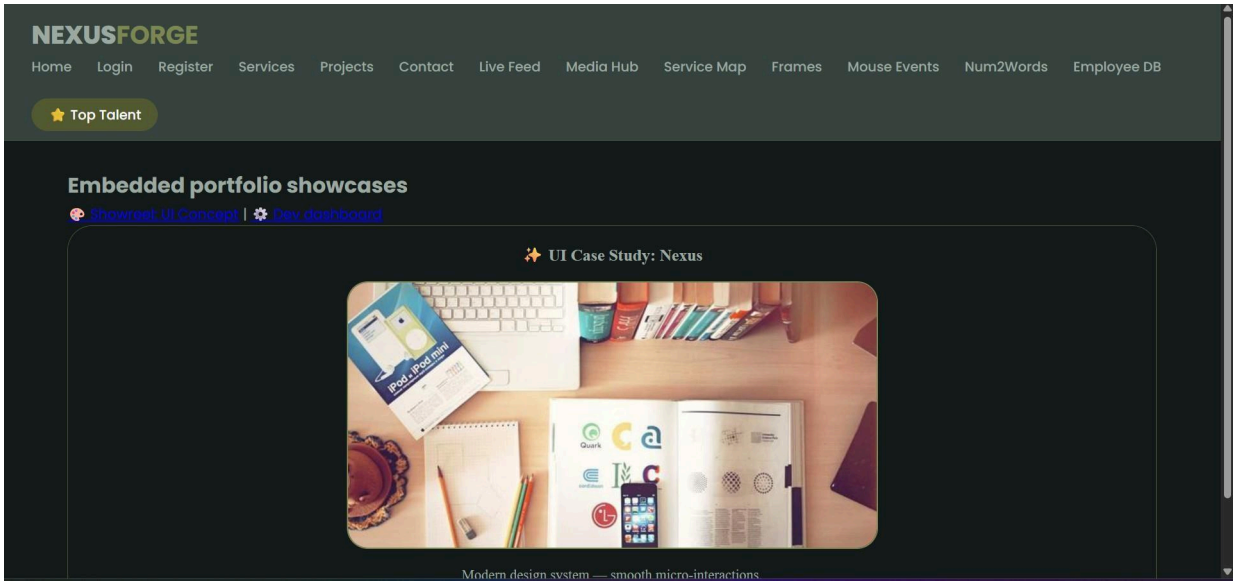
data.txt – contains pipe-separated freelancer records (name, expertise, rating).

data.json – contains an array of objects with name, service type, and rating.

Users can click dedicated buttons to load and display this data as styled cards. The module also loads the JSON data automatically when the page loads. This improves user experience by showing fresh content instantly.

5.2.5 Embedded Media Module (Iframe with Images)

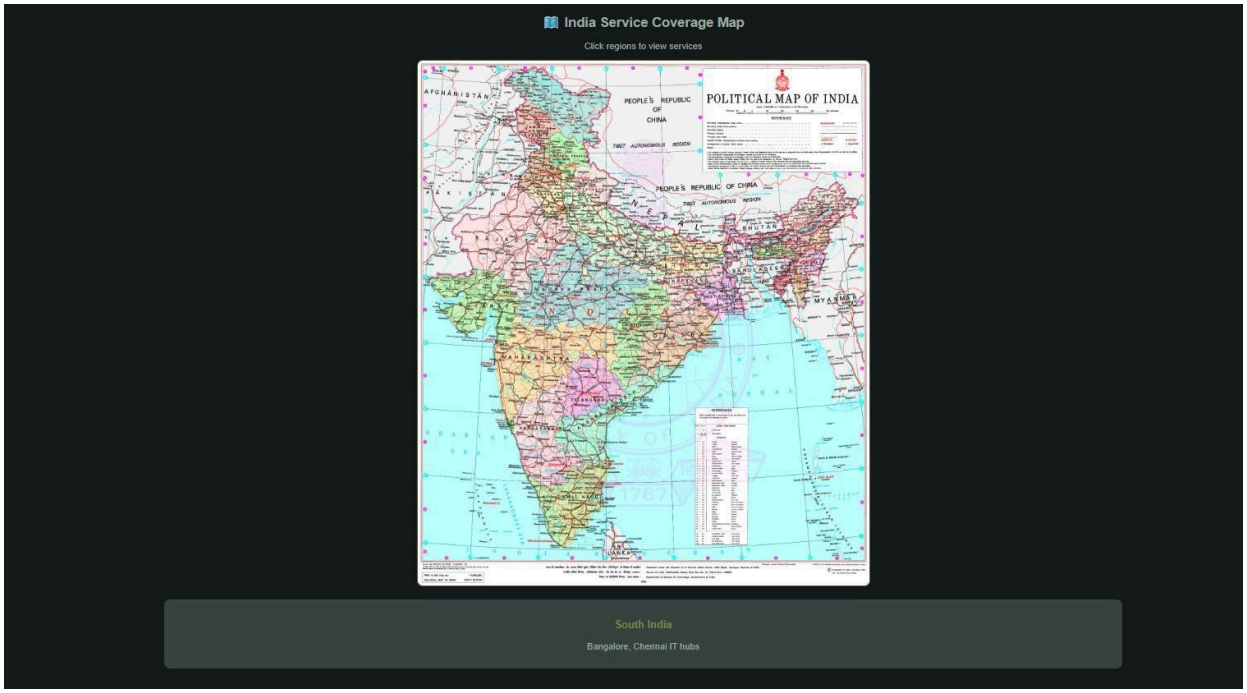
The Embedded Media Module allows the system to display external HTML pages inside an iframe. Two anchor links are provided: one loads a design concept page (design-show.html) and the other loads a developer dashboard page (dev-demo.html). Both pages contain an
<img> tag showing relevant images.



This feature improves content richness by embedding visual media directly within the main page. Users can switch between different image-based pages without leaving the current page.

5.2.6 Interactive Map Module (Image Map with Hotspots)

The Interactive Map Module enables users to explore region-wise freelance expertise by clicking on hotspots over an India map image. The map uses an <img> tag with a <map> element defining five polygonal <area> regions: North, West, South, East, and Central India.

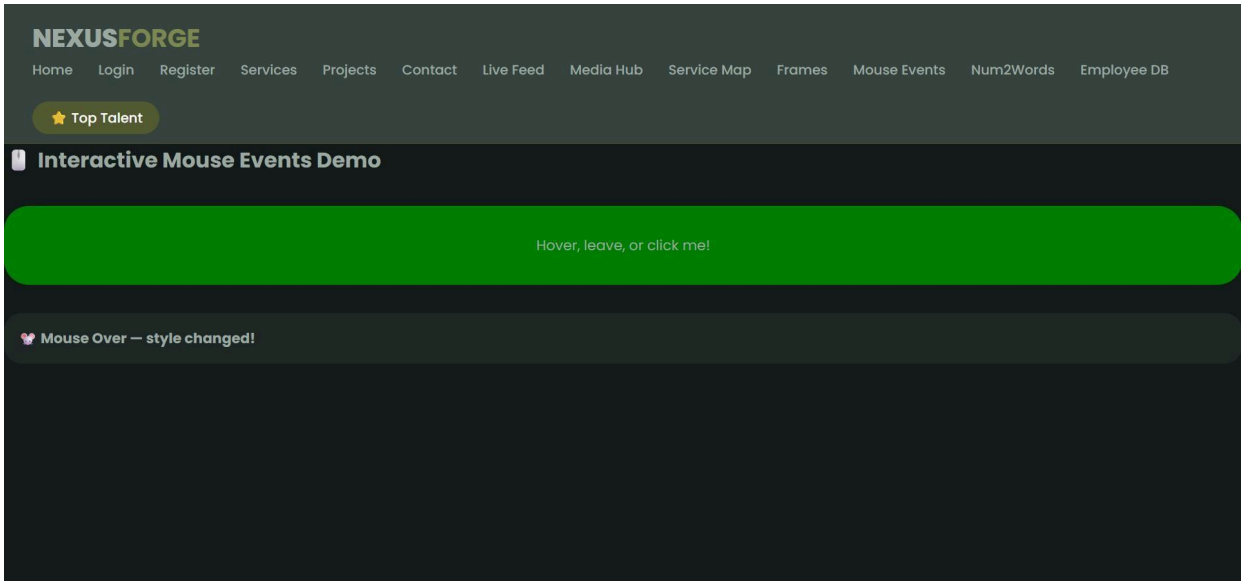


When a user clicks a region, a JavaScript function updates an information panel below the map with a description of freelance specialisations available in that region. This module makes service discovery intuitive and engaging.

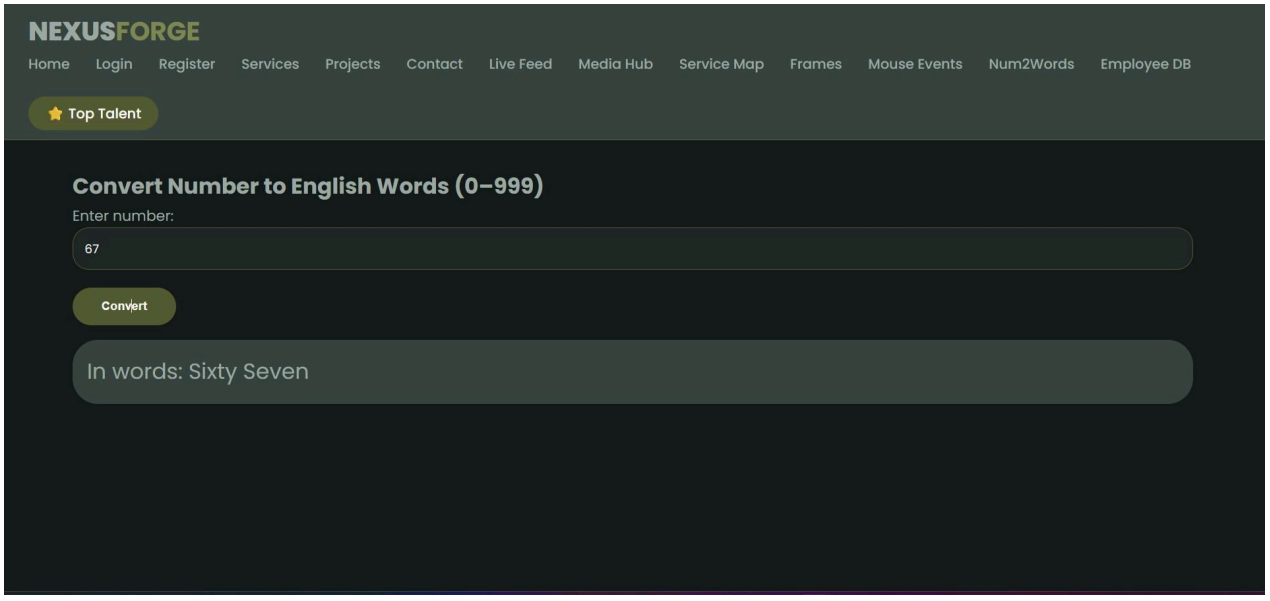
5.2.7 Additional Demonstration Modules

To fulfil all laboratory requirements, the system includes several small demonstration modules:

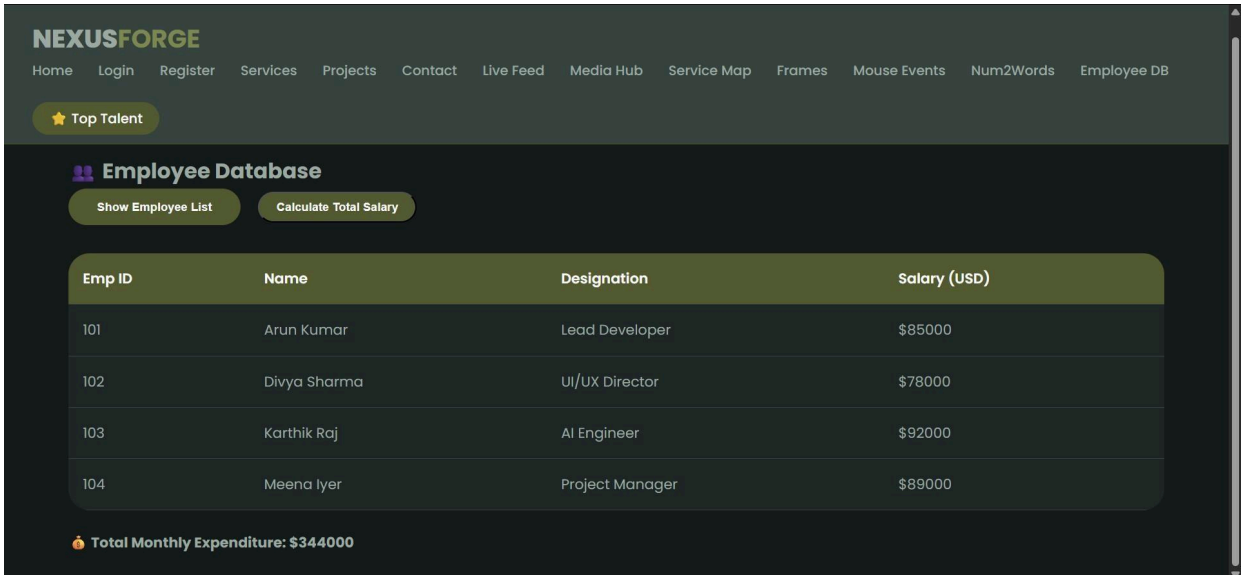
Mouse Events Module (mouse_events.html) – A box that changes background colour and displays messages on mouseover, mouseout, and click events.



Number-to-Words Module (number_words.html) – Converts a numeric input (0-999) into English words. It validates input to reject letters, special characters, and numbers greater than 999.



Employee Object Array Module (employee.html) – Stores an array of employee objects (id, name, designation, salary). Buttons allow users to display the employee table and calculate the total salary expenditure.



Top Talent Animation Module (top_talent.html) – Displays a table of top freelancers with CSS keyframe animations (slide-in and text glow).

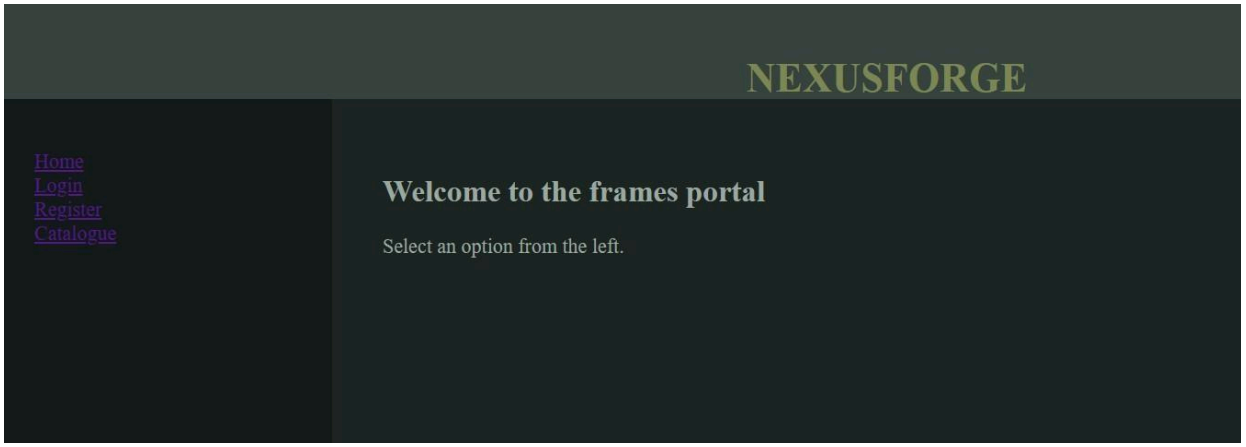
5.2.8 Frameset Layout Module

The Frameset Layout Module demonstrates a legacy three-frame structure using <frameset> instead of <body>. It consists of:

Top frame – website banner.

Left frame – navigation links.

Right frame – content area that loads login, registration, or catalogue pages when links are clicked.



This module satisfies the frames experiment from the laboratory manual while maintaining a professional freelance theme.

CHAPTER 6

RESULTS AND DISCUSSION

6.1 Results

The Online Freelance Services Website was successfully developed and tested across modern web browsers. All pages function as intended and present a consistent user interface aligned with a professional freelance platform.

The home page displays the main interface, including service highlights, process steps, and a structured table of freelancer ratings. The login page provides a basic authentication interface with user feedback through alerts. The registration page includes a complete form with
real-time validation for all required fields.

The services page presents available freelance offerings using structured cards along with supporting lists. The projects page displays sample freelance projects along with a table containing project-related details such as client name, service type, and delivery status.

The AJAX module demonstrates dynamic data loading from both text and JSON files without page refresh. The loaded data is displayed as freelancer profiles and project-related information, simulating a live freelance marketplace feed.

The embedded content module uses an iframe to display different pages containing design previews and development outputs. The map module includes an interactive India map, where users can click on different regions to view region-specific freelance expertise.

Additional modules demonstrate JavaScript functionalities. The mouse events page responds to user interaction by changing styles and displaying messages. The number-to-words module
converts numeric input into words with proper validation. The employee module displays structured data and calculates total salary expenditure. The top talent page includes animated tables to highlight freelancer rankings. The frames module demonstrates a multi-frame layout for navigation and content display.

6.2 Performance Evaluation

The system performs efficiently under normal usage conditions. Since it is a client-side application, page load times are minimal and data retrieval using AJAX is fast due to local file access.

The use of standard web technologies ensures smooth rendering across browsers. Animations and interactive elements function without noticeable delay, providing a responsive user experience. The absence of a backend reduces processing overhead and simplifies execution.

From a resource perspective, the system maintains a compact structure with multiple pages, reusable styles, and modular JavaScript functions. This contributes to maintainability and scalability for future enhancements.

6.3 Resource Utilization

The system was tested using multiple scenarios to verify functionality, validation, and user interaction. Test cases were designed for form validation, dynamic data loading, interactive modules, and navigation features.

All validation rules in the registration form were successfully enforced, including constraints on name, email, password, and phone number. AJAX operations correctly fetched and displayed
data from both text and JSON sources. The interactive map responded accurately to user clicks, updating region-specific information.

The iframe module successfully loaded different pages based on user input. The number-to-words converter handled valid and invalid inputs appropriately. The employee module displayed data correctly and computed total salary values without errors. Mouse events and animation features performed as expected.

All test cases were executed successfully, confirming that the system meets the specified functional requirements.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 Conclusion

The Online Freelance Services Website has been successfully developed as a complete web application that integrates all required web technology concepts into a single platform. The system demonstrates the practical application of HTML, CSS, and JavaScript through a structured and functional freelance marketplace.

The project includes multiple modules such as authentication, service display, project showcase, dynamic data loading, interactive mapping, and embedded content. It also incorporates essential features like form validation, event handling, and data processing using JavaScript.

By transforming academic requirements into a professional freelance platform, the system provides both educational value and practical relevance. It is lightweight, easy to deploy, and does not require backend support, making it suitable for demonstration and learning purposes.

Overall, the system achieves its objective of combining all laboratory experiments into a unified application while maintaining usability and clarity.

7.2 Future Work

Although the system meets current requirements, several enhancements can be implemented to improve functionality and scalability.

Future developments may include integrating a backend system to store user data and manage interactions more effectively. Real-time communication features can be added to enable direct interaction between clients and freelancers. Payment integration can be introduced to support transaction-based services.

Additional improvements may include user profile customization, portfolio uploads, and enhanced analytics for tracking freelance performance. Deployment on cloud platforms with domain integration can further extend accessibility.

These enhancements would transform the system into a more advanced and production-ready freelance platform.

REFERENCES

[1] M. D. Han, "A Systematic Review of Freelance Platforms: Opportunities and Challenges," International Journal of Web & Semantic Technology, vol. 12, no. 3,
pp. 1-15, 2021.

[2] R. Fielding, J. Gettys, J. Mogul, et al., "Hypertext Transfer Protocol – HTTP/1.1," IETF RFC 2616, 1999. [Online].
Available: https://www.ietf.org/rfc/rfc2616.txt

[3] T. Berners-Lee, "HyperText Markup Language (HTML)," World Wide Web Consortium (W3C), 2014. [Online]. Available: https://www.w3.org/html

[4] B. Bos, H. W. Lie, C. Lilley, and I. Jacobs, "Cascading Style Sheets, Level 2 (CSS2) Specification," W3C Recommendation, 1998. [Online].
Available: https://www.w3.org/TR/CSS2/

[5] D. Flanagan, JavaScript: The Definitive Guide, 7th ed. O'Reilly Media, 2020.

[6] MDN Web Docs, "Fetch API – Web APIs," Mozilla Developer Network. [Online]. Available:
https://developer.mozilla.org/en-US/docs/Web/API/Fetch_API

[7] Google Fonts, "Poppins – Google Fonts," 2025. [Online].
Available: https://fonts.google.com/specimen/Poppins

[8] Wikimedia Commons, "India administrative map blank.svg," 2020. [Online]. Available:
https://commons.wikimedia.org/wiki/File:India_administrative_map_blank.svg

[9] K. K. Delabaty and M. I. Al-Turaiki, "Automated Attendance Management Systems: A Systematic Literature Review," Education and Information Technologies, vol. 27, no. 1, pp. 1-24, 2022.

[10] J. Resig, Pro JavaScript Techniques, 2nd ed. Apress, 2015.

[11] E. Robson and E. Freeman, Head First HTML and CSS, 2nd ed. O'Reilly Media, 2012.

APPENDICES

Appendix 1: Source Code Pseudocode

System Overview

The Online Freelance Services Website connects freelancers and clients using HTML5, CSS3 (inline/internal/external), JavaScript, and AJAX. Features include: registration/login with validation, dynamic fetching from .txt and .json files, interactive India map with hotspots, iframe with images, mouse events, number-to-words conversion, employee salary array, and frameset layout.

System Initialization

```
START
  Load external CSS and Google
Fonts On page load: attach event
listeners On data.html: call
loadFromJson() END
```

Registration Validation

```
FUNCTION
  validateRegistration() name =
  get('fullname').value
  IF not /^[A-Za-z]{6,}$/.test(name) THEN alert("Name: min 6 alphabets"); RETURN false
  password = get('regPassword').value
  IF password.length < 6 THEN alert("Password min 6 chars"); RETURN
  false email = get('regEmail').value
  IF not emailRegex.test(email) THEN alert("Valid email required"); RETURN false
  phone = get('phone').value
  IF not /^\d{10}$/.test(phone) THEN alert("Phone: 10 digits"); RETURN
  false IF no gender radio checked THEN alert("Select gender"); RETURN
  false
  IF get('serviceCat').value == "" THEN alert("Select service category"); RETURN
  false IF get('address').value == "" THEN alert("Address required"); RETURN false
  alert("Form validated successfully")
  RETURN
true END
FUNCTION
```

AJAX from TXT

```
FUNCTION
  loadFromTxt()
  fetch('data.txt')
    .then(res => res.text())
    .then(data => {
      lines =
      data.split('\n')
      for each line:
        [name, expertise, rating] = line.split('|')
```

```
      create card and append to #txtOutput
    })
END FUNCTION
```

AJAX from JSON

```
FUNCTION
  loadFromJson()
  fetch('data.json')
    .then(res => res.json())
    .then(items => {
      for each item: create card with name, serviceType, rating
    })
END FUNCTION
```

Map Hotspot

```
FUNCTION showRegion(regionName, description)
  infoDiv.innerHTML = "<div> 📍 " + regionName + "</div><p>" + description + "</p>"
  infoDiv.scrollIntoView()
END FUNCTION
```

Number-to-Words (0-999)

```
FUNCTION
  numberToWords(num) if
  num==0 return "Zero"
  word = ""
  if num>=100: word += ones[floor(num/100)] + " Hundred "; num %=
  100 if num>=10 && num<=19: word += teens[num-10]; num=0
  else if num>=20: word += tens[floor(num/10)] + " "; num %=
  10 if num>0: word += ones[num]
  return
word.trim()
END
FUNCTION
```

Employee Array & Salary

```
employees = [ {id:101,name:"Arun",desig:"Lead Dev",salary:85000}, ... ]
FUNCTION displayEmployees()
  build HTML table from employees array
END FUNCTION

FUNCTION
  calculateTotalSalary() total
  = sum of salaries
  display
total END
FUNCTION
```

System Features Implemented

- Secure registration/login with validation
- AJAX from .txt and .json (no page reload)
- India map hotspots (North, West, South, East, Central)
- Iframe that loads image-containing pages
- CSS keyframe animations (fadeUp, fadeSlide, glowText)
- Mouse events (hover, leave, click) with style changes
- Number-to-words converter (0-999, rejects invalid input)
- Employee object array with salary total
- Frameset layout (three-frame)
- Dark-theme colour palette (#151a1a, #38443f, #545b33, #7f8954, #9eaba3)

Appendix 2: System Test Verification

Test Case	Description	Result
TC-01	Registration – name <6 letters	Alert → PASS
TC-02	Registration – phone not 10 digits	Alert → PASS
TC-03	Registration – all valid fields	Alert → PASS
TC-04	AJAX TXT – load and display cards	4 cards displayed → PASS
TC-05	AJAX JSON – load and display cards	4 cards displayed → PASS
TC-06	Map – click on South India	Panel shows Bangalore/AI info → PASS
TC-07	Iframe – click UI Concept	loads design-show.html with image → PASS
TC-08	Number-to-words – input 45	displays "Forty Five" → PASS
TC-09	Number-to-words – input 1000	alert "0-999 only" → PASS
TC-10	Employee – show table	4 employees listed → PASS
TC-11	Employee – total salary	₹344,000 displayed → PASS
TC-12	Mouse events – hover	background changes → PASS
TC-13	Keyframe animation – top_talent.html	slide-in and glow → PASS
TC-14	Frameset – left navigation loads right frame	content changes → PASS

Final System Statistics

Metric	Value
Total HTML pages	14
JavaScript functions	20+
Keyframe animations	4
AJAX data sources	2 (TXT + JSON)
Map hotspots	5
Form validated fields	8
Test cases passed	14/14

Final System Status

The Online Freelance Services Website has been fully developed and successfully tested. It integrates all required experiments (lists, tables, forms, validation, AJAX from TXT/JSON, image map, iframe, inline/internal/external CSS, keyframes, mouse events, number-to-words, employee array, frameset) into a single, professional freelance platform. The system is responsive, follows the specified dark-theme colour palette, and is ready for deployment.

simplilearn SkillUP

CERTIFICATE OF COMPLETION

Nandana Pramod

has successfully completed the online course:

Java Servlet Basics and JSP 101

This professional has demonstrated initiative and a commitment to deepening their skills and advancing their career. Well done!

1st May 2026

Certificate code : 10173878

Krishna Kumar

CEO, Simplilearn

LinkedIn Learning

HTML, CSS, and JavaScript: Building the Web

Course completed by Nandana Pramod

Apr 30, 2026 at 07:27AM UTC • 3 hours 37 minutes

Top skills covered

Web Development | Cascading Style Sheets (CSS) | HTML

Shea Hanson, Head of Learning Content Strategy

Certificate ID: ff43afdf5be0bb6140407192d039c759383d3a2c2c8c9d966cd3652973206f93

COURSE COMPLETION